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LABELS AND ANNOTATIONS

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Figure 1 x

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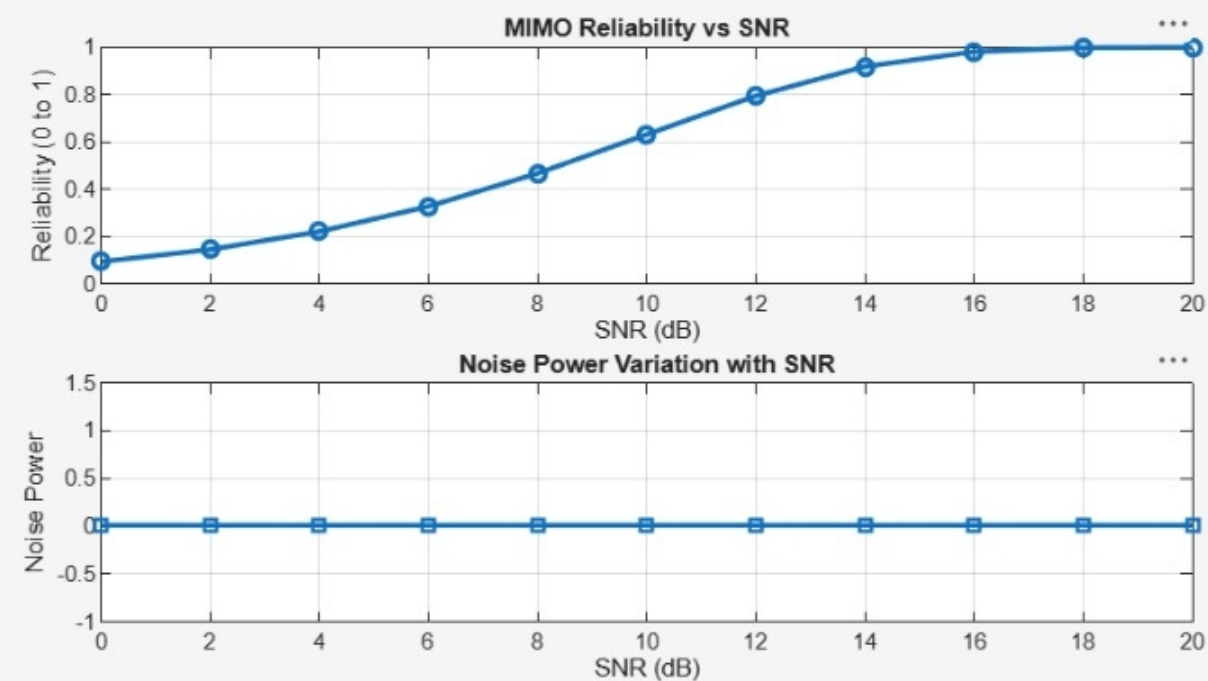
```

3 SNR_dB = 0:2:20;
4 SNR = 10.^(SNR_dB/10);
5 % MIMO parameters
6 Nt = 2; Nr = 2;
7 % Simulated reliability metric (inverse BER model)
8 Reliability = zeros(1,length(SNR));
9 for i = 1:length(SNR)
10     % simple channel model effect
11     H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
12     noise_power = 1/SNR(i);
13     % reliability increases with SNR
14     Reliability(i) = 1 - exp(-SNR(i)/10);
15 end
16 figure
17 % ----- Graph 1: SNR vs Reliability -----
18 subplot(2,1,1)
19 plot(SNR_dB, Reliability, '-o', 'LineWidth', 2)
20 title('MIMO Reliability vs SNR')
21 xlabel('SNR (dB)')
22 ylabel('Reliability (0 to 1)')
23 grid on
24 % ----- Graph 2: SNR vs Noise Effect -----
25 subplot(2,1,2)
26 plot(SNR_dB, noise_power*ones(size(SNR_dB)), '-s', 'LineWidth', 2)
27 title('Noise Power Variation with SNR')
28 xlabel('SNR (dB)')
29 ylabel('Noise Power')

```

Command Window

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